

Quantum Annealing Algorithms for Estimating Ising Partition Functions

Haowei Li^{1,2,3}, Zhiyuan Yao^{4,*}, and Xingze Qiu^{1,†}

¹*School of Physics Science and Engineering, Tongji University, Shanghai 200092, China*

²*Laboratory of Quantum Information, University of Science and Technology of China, Hefei 230026, China*

³*Institute for Advanced Study, Tsinghua University, Beijing 100084, China*

⁴*Lanzhou Center for Theoretical Physics, Key Laboratory of Theoretical Physics of Gansu Province, Key Laboratory of Quantum Theory and Applications of MoE, Gansu Provincial Research Center for Basic Disciplines of Quantum Physics, Lanzhou University, Lanzhou 730000, China*

 (Received 5 May 2025; revised 27 January 2026; accepted 11 February 2026; published 12 March 2026)

Estimating partition functions of Ising spin glasses is a cornerstone of statistical physics and computational science, yet it remains classically challenging due to its $\#P$ -hard complexity. While Jarzynski's equality offers a theoretical pathway, its practical application is crippled at low temperatures by rare, divergent statistical fluctuations. Here, we introduce a quantum protocol that overcomes this fundamental limitation by synergizing reverse quantum annealing with optimized nonequilibrium initial distributions. Our method dramatically suppresses the estimator variance, achieving saturation in the low-temperature regime where existing methods fail. Numerical benchmarks on the Sherrington-Kirkpatrick spin glass and the 3-SAT problem demonstrate that our protocol reduces computational scaling exponents by over an order of magnitude (e.g., from ~ 8.5 to ~ 0.5), despite retaining exponential system-size dependence. Crucially, our protocol circumvents stringent adiabatic constraints, making it feasible for near-term quantum devices like superconducting qubits, trapped ions, and Rydberg atom arrays. This Letter provides a methodological framework for quantum-enhanced estimation in spin glass thermodynamics and beyond by harnessing nonadiabatic quantum dynamics to address a classically difficult problem.

DOI: [10.1103/8gmb-p619](https://doi.org/10.1103/8gmb-p619)

Introduction—Ising spin glasses (ISGs) play a pivotal role in both fundamental research and practical applications across diverse disciplines [1–4], including statistical physics [5], combinatorial optimization [6], and machine learning [7]. Despite its broad relevance, a longstanding challenge persists: estimating the Ising partition functions (IPFs) of these complex systems [8], which is classically difficult, being $\#P$ -hard in the worst case [9]. Conventional approaches to this problem face fundamental roadblocks, particularly in the physically interesting low-temperature regime.

The Jarzynski's equality (JE) [10–15], an elegant identity from nonequilibrium statistical mechanics, theoretically connects the free energy difference between two systems to the average work done during a transformation between them. However, its practical utility for IPF estimation is crippled at low temperatures by rare events dominating the exponential average [16,17]. Classical methods based on Markov-chain Monte Carlo (MCMC) [18,19] can exhibit extremely long equilibration and autocorrelation times in rugged low-temperature energy landscapes, resulting in trapping within metastable basins. While enhanced sampling

methods offer partial mitigation [20–23], low-temperature barriers remain a persistent bottleneck in spin-glass regimes due to rapidly increasing mixing and round-trip times [24]. In view of this fundamental barrier, advanced quantum algorithms have emerged as promising candidates for circumventing classical limitations. Recent proposals include quantum MCMC [25], quantum phase estimation [26], and deterministic quantum computation with one clean qubit (DQC1)-based algorithms [27,28]. However, these approaches remain largely confined to idealized settings and demand quantum resources beyond what is currently feasible on noisy intermediate-scale quantum (NISQ) devices [29].

Here, we introduce a new algorithmic paradigm that is explicitly designed to overcome these barriers by harnessing the native physics of NISQ hardware. As illustrated in Fig. 1, our approach is a synergistic hybrid algorithm that combines a classically optimized, nonequilibrium initial state distribution with a quantum evolution generated by **reverse quantum annealing (RQA)** [30–34]. The choice of RQA is not one of convenience but is fundamental to the protocol's performance and feasibility. First, RQA is a hardware-native physical process on leading quantum annealing platforms, minimizing control overhead. Second, RQA excels at the local refinement of known

*Contact author: yaozy@lzu.edu.cn

†Contact author: xingze@tongji.edu.cn

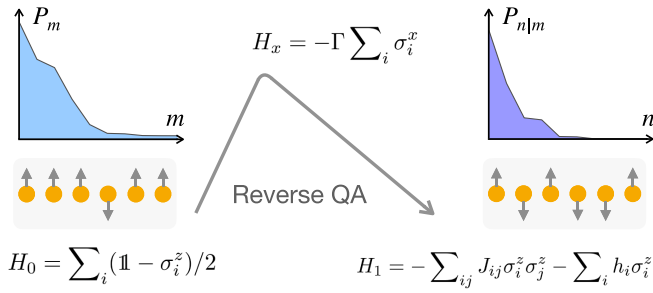


FIG. 1. Schematics of our protocol for estimating IPFs. The protocol begins by preparing the system in an initial state sampled from a distribution P_m . Unlike conventional JE approaches that assume thermal equilibrium (e.g., Gibbs distributions), P_m is intentionally designed to deviate from equilibrium to suppress the variance. The system then goes through a nonequilibrium RQA process [Eq. (1)], in which the Hamiltonian transitions from a trivial free model H_0 to the desired target ISG H_1 with the aid of a driver H_x . Finally, projective measurements in the eigenbasis of H_1 yield outcomes governed by the conditional probability $P_{n|m}$ [see details above Eq. (2)]. These outcomes are processed via the estimator to reconstruct the IPFs of H_1 [Eq. (3)].

classical states, making it the ideal quantum engine to explore the promising regions of the solution space identified by our classical optimization. Third, and most profoundly, our protocol operates optimally in a nonadiabatic regime. An infinitely long, perfectly adiabatic evolution is not only unnecessary but suboptimal. This means that the primary constraint of NISQ hardware—limited coherence time—is not a fatal flaw for our protocol but is instead aligned with its optimal operating regime. Our protocol can thus be implemented on current platforms like superconducting qubits [35–37], trapped ions [38,39], and Rydberg atom arrays [40,41].

Our Letter demonstrates two key advances. First, the protocol completely circumvents the rare-event bottleneck of JE-based approaches, achieving saturation of the estimator variance at low temperatures and reducing the computational scaling exponent by over an order of magnitude. Second, our protocol positions quantum annealing (QA) as a versatile tool for estimating IPFs, which extends conventional QA’s utility beyond the ground-state search of ISGs [42–44]. This enriches the research that utilizes QA-enabled Gibbs sampling [45–47] in approximating the finite-temperature properties of ISGs. By exploiting quantum dynamics to sidestep the critical failures of existing methods, our protocol establishes a practical and powerful application for near-term quantum processors in tackling classically difficult problems. This Letter advances applications in ISG physics, optimization, and machine learning, where IPFs estimation is critical yet classically difficult.

Reverse quantum annealing protocol—RQA enables state-selective initialization of QA dynamics—a feature critical for applications such as hybrid quantum-classical optimization [31] and quantum simulations [32]. The RQA

protocol can be described by the following form of a time-dependent Hamiltonian [33,34]

$$H(t) = s(t)H_1 + [1 - s(t)]\lambda(t)H_x + [1 - s(t)][1 - \lambda(t)]H_0. \quad (1)$$

Here, the time-dependent parameters $s(t)$ and $\lambda(t)$ both grow from 0 to 1 over the course of the unitary time evolution $U(\tau) = \mathcal{T} \exp[-i \int_0^\tau H(t)dt]$ with τ the total time and $\mathcal{T}$ the time-ordering operator. The system thus transitions from the initial Hamiltonian H_0 to the target Hamiltonian H_1 , aided by the driver Hamiltonian H_x . In our context of the ISG problem, $H_0 = \sum_{i=1}^N (1 - \sigma_i^z)/2$, $H_1 = -\sum_{i,j=1}^N J_{ij} \sigma_i^z \sigma_j^z - \sum_{i=1}^N h_i \sigma_i^z$, and $H_x = -\Gamma \sum_{i=1}^N \sigma_i^x$ where 1 is the identity operator, $\{\sigma_i^{x/z}\}$ are the Pauli- x/z operators, and N is the system size. The spin-spin couplings J_{ij} and the local fields h_i defines a model instance [42,43], and Γ sets the scale of quantum fluctuations for state transitions. Hereafter, we set $\hbar = 1$ and $k_B = 1$, use Γ as our energy unit, and choose $s(t) = \lambda(t) = t/\tau$. The choice of H_0 is physically motivated, as its eigenstates are computational basis states that can be efficiently prepared on a wide range of quantum hardware platforms, making RQA the natural framework for initiating the desired quantum evolution.

Let $\{E_m^0, |\psi_m^0\rangle\}$ and $\{E_n^1, |\psi_n^1\rangle\}$ denote the ascending eigenvalues and corresponding eigenstates of H_0 and H_1 , respectively. The system is initialized by randomly sampling $|\psi_m^0\rangle$ according to a sampling function P_m , which satisfies $P_m > 0$ and $\sum_{m=1}^D P_m = 1$ with $D = 2^N$ the dimension of the Hilbert space. Under the RQA dynamics, the initial state evolves to $|\psi(\tau)\rangle = U(\tau)|\psi_m^0\rangle$. The following projective measurement in the eigenbasis of H_1 yields a trajectory $|\psi_m^0\rangle \rightarrow |\psi_n^1\rangle$ with a conditional probability $P_{n|m} = |\langle \psi_n^1 | U(\tau) | \psi_m^0 \rangle|^2$. Because of the unitarity of $U(\tau)$, we have $\sum_{m=1}^D P_{n|m} = 1$. Now we are ready to reconstruct the IPFs of H_1 as

$$Z_1(\beta) = \sum_{n=1}^D \exp(-\beta E_n^1) = \sum_{m,n=1}^D \frac{\exp(-\beta E_n^1)}{P_m} P_{n|m} P_m, \quad (2)$$

where $\beta = 1/T$ is the inverse temperature. This estimator leverages correlations between initial (P_m) and transition ($P_{n|m}$) probabilities to bypass direct summation over D states.

Note that choosing $P_m = P_m^G(\beta) \equiv \exp(-\beta E_m^0)/Z_0(\beta)$ (Gibbs distribution) will reduce Eq. (2) to the celebrated JE: $Z_1(\beta)/Z_0(\beta) = \exp(-\beta \Delta F) = \langle \exp(-\beta W) \rangle$ [10–13]. Here, $Z_0(\beta) = [1 + \exp(-\beta)]^N$ is the trivial partition function of H_0 , $\Delta F = -\beta^{-1} \log[Z_1(\beta)/Z_0(\beta)]$ is the free

energy difference, and the expectation value $\langle \exp(-\beta W) \rangle = \sum_{m,n=1}^D \exp(-\beta W_{nm}) P_{n|m} P_m$ with $W_{nm} = E_n^1 - E_m^0$ the quantum work defined through a two-point measurement scheme [11–13]. By enabling efficient preparation of initial Gibbs ensembles, this protocol establishes quantum processors as practical tools for ISG thermodynamics—directly supporting applications such as validating fluctuation theorems in many-body regimes [48] and estimating free energy for disordered systems [49].

In the following, we focus on estimating the IPFs of H_1 based on Eq. (2), to achieve contrasted protocol advantages over conventional JE-based approaches. To demonstrate the efficiency of our protocol, we benchmark it against two canonical ISG models: the Sherrington-Kirkpatrick (SK) spin glass [50,51] and the random 3-SAT [52]. The Hamiltonian of the SK model reads $H_{\text{SK}} = (1/\sqrt{N}) \sum_{(j \neq i)=1}^N J_{ij} \sigma_i^z \sigma_j^z + \sum_{i=1}^N h_i \sigma_i^z$, where J_{ij} and h_i are independent variables sampled from the standard normal distribution. And the random 3-SAT is a fundamental Boolean satisfiability problem. The hard instances are generated via a physics-inspired protocol with planted solutions, ensuring controlled benchmarking in classically difficult regimes [53].

Optimized sampling function—In practice, the convergence of any sampling-based estimator is determined by its variance. The number of samples M_s required to achieve a desired relative error ε scales as $M_s \propto \sigma^2/(\varepsilon^2 Z_1^2)$, where σ^2 is the variance of the estimator. Therefore, minimizing this variance is equivalent to minimizing the total computational cost. The JE estimator suffers from notoriously slow convergence precisely because its variance is large, especially at low temperatures [16,17]. To address this, we propose replacing the conventional Gibbs initial distribution with a tailored nonequilibrium one. As shown below, this approach dramatically suppresses estimator variance, especially in the low-temperature regime.

From Eq. (2), the IPFs of H_1 can be estimated as

$$Z_1(\beta) = \langle z_{m,n}(\beta) \rangle \approx Z_{\text{est}}(\beta) \equiv \frac{1}{M_s} \sum z_{m,n}(\beta), \quad (3)$$

where $\langle \cdot \rangle$ denotes averaging over trajectories $|\psi_m^0\rangle \rightarrow |\psi_n^1\rangle$ sampled with the probability $P_{n|m} P_m$, and $z_{m,n}(\beta) = \exp(-\beta E_n^1)/P_m$ is the corresponding random variable. While the equality holds exactly as $M_s \rightarrow \infty$, finite sampling necessitates clever design of the initial distribution P_m and transition probability $P_{n|m}$ to minimize estimator variance. Our protocol thus operates as a hybrid classical-quantum algorithm, i.e., the initial state is randomly chosen from P_m , a classical distribution designed to minimize statistical fluctuations in Z_{est} ; the transition $|\psi_m^0\rangle \rightarrow |\psi_n^1\rangle$ is governed by $P_{n|m}$, which is generated by the quantum dynamics of RQA. For the latter, it is known that increasing the evolution time τ can reduce the required

M_s [65]. However, practical limitations—notably finite coherence time in quantum hardware—constrain τ , demanding a balance between quantum resource allocation and sampling efficiency. This raises the central question: what choice of P_m minimizes the variance of Z_{est} for a given RQA protocol with finite τ ?

To address this problem, we examine the variance of the estimator $z_{m,n}(\beta)$, defined as [53]

$$\begin{aligned} \sigma^2(\beta) &= \sum_{m,n=1}^D [z_{m,n}(\beta) - Z_1(\beta)]^2 P_{n|m} P_m \\ &= \sum_{m=1}^D \frac{\mu_m(\beta)}{P_m} - Z_1^2(\beta). \end{aligned} \quad (4)$$

Here, $\mu_m(\beta) = \sum_{n=1}^D \exp(-2\beta E_n^1) P_{n|m}$. Guided by importance sampling principles [66], we seek to minimize $\sigma^2(\beta)$ by optimizing P_m . A direct constrained minimization yields the theoretically optimal distribution $P_m = \mathcal{P}_m(\beta) \equiv \sqrt{\mu_m(\beta)}/\mathcal{N}_{\mathcal{P}}(\beta)$, where $\mathcal{N}_{\mathcal{P}}(\beta)$ ensures normalization [53]. While $\mathcal{P}_m(\beta)$ guarantees the minimal variance, its direct computation is infeasible for finite τ in practical implementations. To resolve this, we develop an efficient approximation scheme for $\mathcal{P}_m(\beta)$ as follows.

The simplest approximation replaces $\mathcal{P}_m(\beta)$ with a Gibbs distribution $P_m^G(\alpha)$, parametrized by a variational inverse temperature α . For a fixed β , we optimize α to minimize $\sigma^2(\beta)$. This reduces to solving the minimization problem $\min_{\alpha} \sum_{m=1}^D \mu_m(\beta)/P_m^G(\alpha)$, and from which we identify the unique optimal α that satisfies [53]

$$\frac{N}{1 + \exp(\alpha)} = \sum_{E=0}^N E \cdot Q_E(\alpha; \beta). \quad (5)$$

Here, $Q_E(\alpha; \beta) \propto \exp(\alpha E) \sum_{m \in I_E} \mu_m(\beta)$ is a distribution, and $I_E = \{m | E_m^0 = E\}$ groups degenerate initial states $|\psi_m^0\rangle$ sharing the same energy E [Fig. 2(a)]. Numerical results for the SK spin glass [Fig. 2(b)] reveal two key features: (i) as β increases, α rises initially before saturating to a τ -dependent value; (ii) longer τ increases α , concentrating $P_m^G(\alpha)$ and reducing the computational effort to estimate $Z_1(\beta)$, which is consistent with previous works [65]. These behaviors starkly contrast with JE, which fixes $\alpha \equiv \beta$. JE's rigidity leads to inefficient sampling in low temperatures, as high-energy states are inadequately probed. Similar trends have been observed in other complex systems, such as the 3-SAT [53], underscoring the generality of this approach.

While $P_m^G(\alpha)$ outperforms $P_m^G(\beta)$, determining α via Eq. (5) for large system sizes remains challenging. This issue can be mitigated based on three key observations. First, $P_m^G(\alpha)$, while simpler, reasonably approximates the optimal distribution $\mathcal{P}_m(\beta) \propto \sqrt{\mu_m(\beta)}$ [53]. Its defining

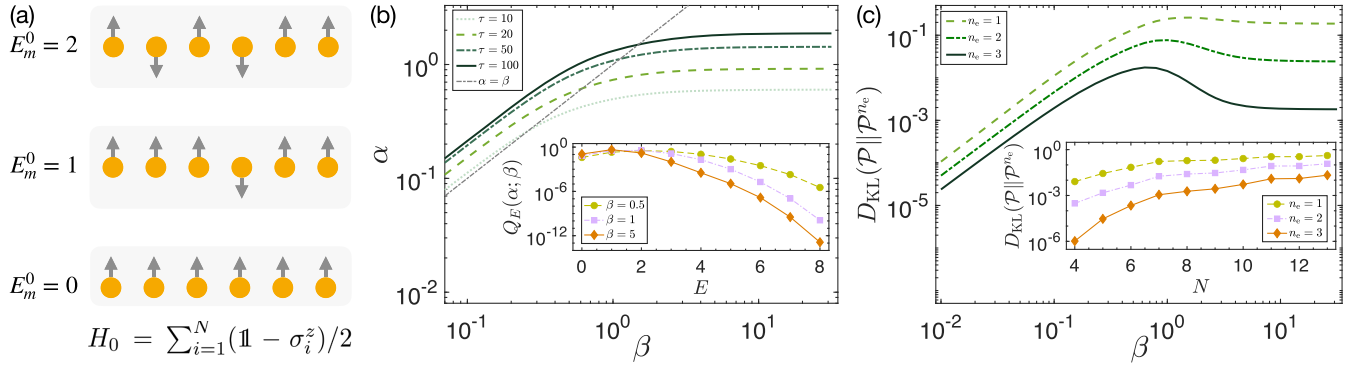


FIG. 2. Approximation strategies for optimized sampling functions in the SK spin-glass model. (a) Initial spin configurations. Arrows represent spins in the eigenstates of σ^z , $\{|\uparrow\rangle, |\downarrow\rangle\}$ (eigenvalues $+1$ and -1 , respectively). The initial product state $|\psi_m^0\rangle = \bigotimes_{i=1}^N |\gamma_i\rangle$ ($|\gamma_i\rangle \in \{|\uparrow\rangle, |\downarrow\rangle\}$), is an eigenstate of H_0 . The number of $|\downarrow\rangle$ in $|\psi_m^0\rangle$ equals its energy E_m^0 , with $I_E = \{m | E_m^0 = E\}$ indexing degenerate states of energy E . (b) Gibbs approximation of $\mathcal{P}_m(\beta)$. The optimized sampling function $\mathcal{P}_m(\beta)$ is approximated by a Gibbs distribution $\mathcal{P}_m^G(\alpha)$ for total evolution times $\tau = 10, 20, 50, 100$. The gray dash-dotted $\alpha = \beta$ line is drawn for reference purposes. The inset shows the distribution $Q_E(\alpha; \beta)$ for $\beta = 0.5, 1, 5$ with $\tau = 100$. Here, we have chosen the system size $N = 8$. (c) Perturbative approximation of $\mathcal{P}_m(\beta)$. The Kullback-Leibler divergence $D_{\text{KL}}(\mathcal{P} \parallel \mathcal{P}^{n_e}) = \sum_m \mathcal{P}_m \log(\mathcal{P}_m / \mathcal{P}_m^{n_e})$ [67] quantifies the difference between $\mathcal{P}_m(\beta)$ and $\mathcal{P}_m^{n_e}(\beta)$ for $n_e = 1, 2, 3$. Main panel: β dependence ($N = 8, \tau = 100$). Inset: system-size scaling ($N = 4 - 13, \tau = 100, \beta = 10$). All data are averaged over 10^3 Hamiltonian instances.

property, $\mathcal{P}_{m \in I_{E+1}}^G(\alpha) = \exp(-\alpha) \mathcal{P}_{m \in I_E}^G(\alpha)$, leads to the approximation

$$\bar{\nu}_{E+1}(\beta) \approx \exp(-\alpha) \bar{\nu}_E(\beta), \quad (6)$$

where $\bar{\nu}_E(\beta) = \sum_{m \in I_E} \sqrt{\mu_m(\beta)} / \binom{N}{E}$ averages $\sqrt{\mu_m(\beta)}$ over the $E_m^0 = E$ degenerate subspace. Second, combining Eq. (6) and the definition of Q_E , we find that Q_E decays exponentially with large E [see Fig. 2(b), inset]. Consequently, only contributions from low-energy subspaces ($E \ll N$) significantly affect the expectation value on the right-hand side of Eq. (5). Third, for low-energy initial states ($m \in I_E, E \ll N$), we compute $\mu_m(\beta) = \sum_{n=1}^D \exp(-2\beta E_n^1) P_{n|m}$ using experimentally sampled trajectories $|\psi_m^0\rangle \rightarrow |\psi_n^1\rangle$ [53]. Crucially, these trajectories are reused to calculate $Z_1(\beta)$ after determining the sampling function $\mathcal{P}_m$, incurring negligible additional computational costs.

Guided by these findings, we propose a family of sampling functions $\mathcal{P}_m^{n_e}(\beta)$ ($n_e = 1, 2, \dots, N$) to approximate $\mathcal{P}_m(\beta)$. The construction involves four main steps as detailed in Supplemental Material [53]. First, we introduce the auxiliary variables $\{\nu_m(\beta)\}_{m=1,2,\dots,D}$ and set $\nu_m(\beta) = \sqrt{\mu_m(\beta)}$ for $m \in I_{E \leq n_e}$. These $\mu_m(\beta)$ can be obtained experimentally via trajectory sampling as mentioned earlier. Second, for $m \in I_{E > n_e}$, we iteratively construct $\nu_m(\beta)$ via the ansatz

$$\nu_{m \in I_{E+1}}(\beta) = \frac{\exp(-\alpha)}{E+1} \sum_{m' \in J_E(m)} \nu_{m'}(\beta), \quad (7)$$

where α is a parameter determined self-consistently, $J_E(m) = \{m' | m' \in I_E, m \in I_{E+1}, D_H(m', m) = 1\}$ contains states $m' \in I_E$ adjacent to m . Here, $D_H(m', m)$ is the Hamming distance (i.e., the number of different spins) between $|\psi_{m'}^0\rangle$ and $|\psi_m^0\rangle$ [Fig. 2(a)]. This ensures consistency with the exponential decay relation in Eq. (6). Third, we approximate $\mu_m(\beta) \approx \nu_m^2(\beta)$ for $m \in I_{E > n_e}$. Substituting all $\mu_m(\beta)$ into Eq. (5) yields a closed-form solution for α . Finally, the n_e th order approximation is given by $\mathcal{P}_m^{n_e}(\beta) \propto \nu_m(\beta)$, normalized over all m . As shown in Fig. 2(c), we find that increasing n_e significantly improves fidelity of $\mathcal{P}_m(\beta)$, particularly at low temperatures. Besides, the approximation remains robust as system size N grows [Fig. 2(c), inset], with significant error reduction even for modest n_e .

Results—To assess the efficacy of our protocol, we employ the relative estimator variance σ^2/Z_1^2 as the primary performance metric, since it directly determines the computational resources required for a given precision. Numerical results for the SK spin glass and 3-SAT models, presented in Fig. 3, demonstrate a profound advantage over the standard JE-based approach. Notably, in the low-temperature regime, the variance of our protocol—even when using the simplified sampling function $\mathcal{P}_m^{n_e=1}(\beta)$ —saturates, whereas the JE-based method exhibits exponential divergence [Figs. 3(a1) and 3(b1), insets]. Given that estimating IPFs is $\#P$ -hard, quantum computers cannot reasonably be expected to solve this task in polynomial time. Performance is therefore quantified by the exponential scaling of the variance with system size, $\sigma^2/Z_1^2 \sim D^\gamma = 2^{\gamma N}$. At a representative low temperature ($\beta = 10$), we analyze this scaling for our protocol versus

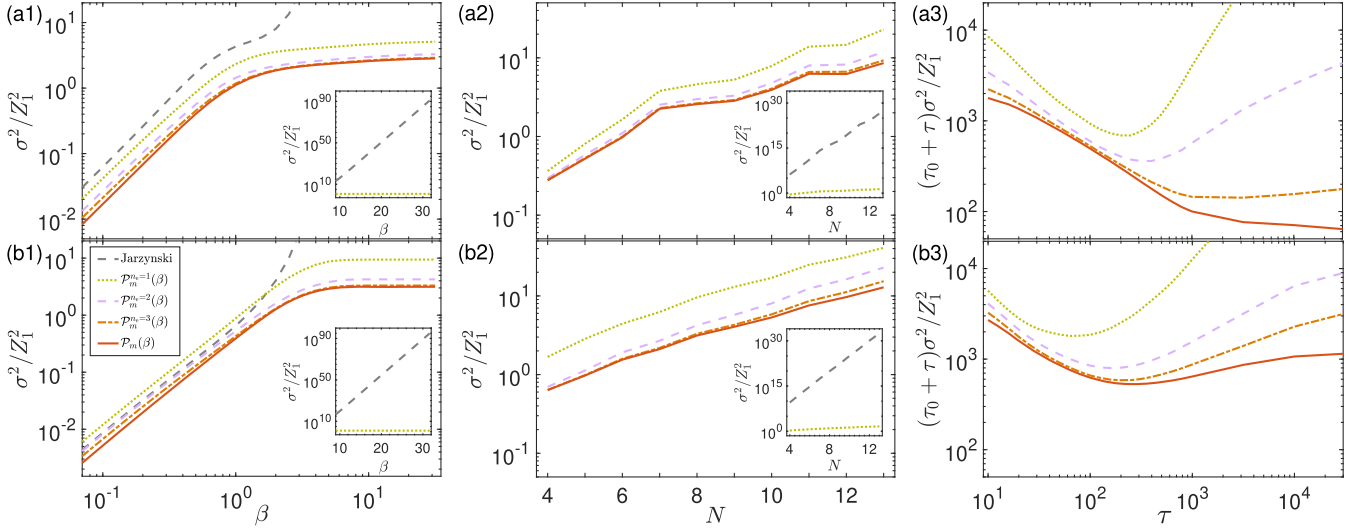


FIG. 3. Performance evaluation of our protocol for the SK spin-glass (a1)–(a3) and 3-SAT (b1)–(b3) models. The red solid line indicates the optimal sampling function $\mathcal{P}_m(\beta)$. And the approximations $\mathcal{P}_m^{n_e}(\beta)$ for $n_e = 1, 2, 3$ are depicted as follows: $n_e = 1$ (light green dotted line), $n_e = 2$ (purple dashed line), and $n_e = 3$ (orange dash-dotted line). Results from the JE-based approach (gray dashed line) are included for comparison. Here, we have chosen the system size $N = 8$ for (a1), (b1), (a3), (b3), the inverse temperature $\beta = 10$ for (a2), (b2), (a3), (b3), and the initialization and measurement time $\tau_0 = 100$ for (a3), (b3). All data are averaged over 10^3 Hamiltonian instances.

the JE baseline [Figs. 3(a2) and 3(b2)]. For the SK spin glass, our protocol with $\mathcal{P}_m^{n_e=1}(\beta)$ achieves a scaling exponent of $\gamma \approx 0.446$, whereas the JE approach yields $\gamma \approx 6.993$ for the hard 3-SAT instances, the exponents are $\gamma \approx 0.502$ for our protocol versus $\gamma \approx 8.451$ for JE. This reduction in the scaling exponent by over an order of magnitude corresponds to a substantial reduction in the exponential scaling of the total computational cost (i.e., the number of experimental runs) required to solve a problem of a given size N . A full complexity analysis of the scaling properties of $\mathcal{P}_m(\beta)$ further corroborates these results [53].

Our protocol is not only powerful but also practical for near-term implementation. Because the number of samples needed scales with the variance, we define the total resource cost as $\mathcal{M}_s = (\tau_0 + \tau)\sigma^2$, where τ_0 denotes the initialization and measurement time per sample. While a longer evolution time τ generally reduces the variance σ^2 , there is a trade-off. As shown empirically in Figs. 3(a3) and 3(b3), this trade-off leads to an optimal, finite τ that minimizes the total cost. This confirms that the most efficient implementation of our protocol operates in a nonadiabatic regime, perfectly aligning with the limited coherence times of NISQ devices. This performance represents a substantial advantage over both the JE baseline and other known approaches in the challenging low-temperature regime. While JE can fail due to rare-event-dominated statistics, classical MCMC methods can become severely hindered by extremely long (in unfavorable cases, effectively exponential) equilibration times in rugged spin-glass landscapes [68,69]; enhanced variants such as annealed importance sampling or generalized-ensemble schemes may

improve exploration but can still struggle with low-temperature barriers [24]. More advanced quantum algorithms based on phase estimation [26] or DQC1 [27,28] require fault-tolerant hardware and are thus impractical for near-term implementation. A detailed complexity comparison is provided in Supplemental Material [53].

Conclusion and discussion—We have introduced an efficient, hardware-native protocol for estimating IPFs, a foundational $\#P$ -hard problem. By synergizing RQA with optimized nonequilibrium initial distributions, our method circumvents the fundamental limitations of conventional approaches based on JE. The protocol's key achievements are the saturation of estimator variance at low temperatures and a substantial improvement in computational efficiency and scalability. These results highlight the power of harnessing nonequilibrium quantum dynamics to tackle classically difficult problems. The underlying algorithmic paradigm—the synergistic use of optimized nonequilibrium initial states and nonadiabatic quantum dynamics—is a generalizable strategy with broad relevance beyond spin glasses. In realistic implementations, environmental noise can substantially affect quantum annealing dynamics [70,71], and a comprehensive noise-aware validation—given ongoing debates over appropriate modeling [72]—is deferred to future work. As quantum hardware matures, such strategies could unlock new insights into the complex energy landscapes of problems in materials science, protein folding, and machine learning [73,74]. Our Letter provides a concrete blueprint for leveraging the unique capabilities of near-term quantum processors [35–41], bridging the gap

between theoretical quantum enhancement and impactful real-world applications.

Acknowledgments—Z. Y. acknowledges support by National Natural Science Foundation of China (12304288, 12247101). X. Q. acknowledges support from the Fundamental Research Funds for the Central Universities and Shanghai Science and Technology project (24LZ1401600).

Data availability—The data that support the findings of this article are openly available [75].

- [1] K. Binder and A. P. Young, Spin glasses: Experimental facts, theoretical concepts, and open questions, *Rev. Mod. Phys.* **58**, 801 (1986).
- [2] H. Nishimori, *Statistical Physics of Spin Glasses and Information Processing: An Introduction* (Oxford University Press, Oxford, 2001).
- [3] D. L. Stein and C. M. Newman, *Spin Glasses and Complexity* (Princeton University Press, Princeton, NJ, 2013).
- [4] P. Charbonneau, E. Marinari, M. Mézard, G. Parisi, F. Ricci-Tersenghi, G. Sicuro, and F. Zamponi, *Spin Glass Theory and Far Beyond* (World Scientific, Singapore, 2023).
- [5] A. D. King *et al.*, Quantum critical dynamics in a 5,000-qubit programmable spin glass, *Nature (London)* **617**, 61 (2023).
- [6] S. Kirkpatrick, C. D. Gelatt, and M. P. Vecchi, Optimization by simulated annealing, *Science* **220**, 671 (1983).
- [7] R. G. Melko, G. Carleo, J. Carrasquilla, and J. I. Cirac, Restricted Boltzmann machines in quantum physics, *Nat. Phys.* **15**, 887 (2019).
- [8] K. Huang, *Statistical Mechanics* (John Wiley & Sons, New York, 1991).
- [9] D. A. Lidar, On the quantum computational complexity of the Ising spin glass partition function and of knot invariants, *New J. Phys.* **6**, 167 (2004).
- [10] C. Jarzynski, Nonequilibrium equality for free energy differences, *Phys. Rev. Lett.* **78**, 2690 (1997).
- [11] H. Tasaki, Jarzynski relations for quantum systems and some applications, [arXiv:cond-mat/0009244](#).
- [12] J. Kurchan, A quantum fluctuation theorem, [arXiv:cond-mat/0007360](#).
- [13] S. Mukamel, Quantum extension of the Jarzynski relation: Analogy with stochastic dephasing, *Phys. Rev. Lett.* **90**, 170604 (2003).
- [14] M. Esposito, U. Harbola, and S. Mukamel, Nonequilibrium fluctuations, fluctuation theorems, and counting statistics in quantum systems, *Rev. Mod. Phys.* **81**, 1665 (2009).
- [15] M. Campisi, P. Hänggi, and P. Talkner, Colloquium: Quantum fluctuation relations: Foundations and applications, *Rev. Mod. Phys.* **83**, 771 (2011).
- [16] C. Jarzynski, Rare events and the convergence of exponentially averaged work values, *Phys. Rev. E* **73**, 046105 (2006).
- [17] G. Xiao and J. Gong, Suppression of work fluctuations by optimal control: An approach based on Jarzynski's equality, *Phys. Rev. E* **90**, 052132 (2014).
- [18] D. Štefankovič, S. Vempala, and E. Vigoda, Adaptive simulated annealing: A near-optimal connection between sampling and counting, *J. ACM* **56**, 1 (2009).
- [19] V. Kolmogorov, A faster approximation algorithm for the Gibbs partition function, in *Proceedings of the 31st Conference on Learning Theory*, edited by S. Bubeck, V. Perchet, and P. Rigollet, Proceedings of Machine Learning Research Vol. 75 (PMLR, Stockholm, Sweden, 2018), pp. 228–249.
- [20] R. M. Neal, Annealed importance sampling, *Stat. Comput.* **11**, 125 (2001).
- [21] B. A. Berg and T. Neuhaus, Multicanonical ensemble: A new approach to simulate first-order phase transitions, *Phys. Rev. Lett.* **68**, 9 (1992).
- [22] F. Wang and D. P. Landau, Efficient, multiple-range random walk algorithm to calculate the density of states, *Phys. Rev. Lett.* **86**, 2050 (2001).
- [23] F. Wang and D. P. Landau, Determining the density of states for classical statistical models: A random walk algorithm to produce a flat histogram, *Phys. Rev. E* **64**, 056101 (2001).
- [24] P. Dayal, S. Trebst, S. Wessel, D. Würtz, M. Troyer, S. Sabhapandit, and S. N. Coppersmith, Performance limitations of flat-histogram methods, *Phys. Rev. Lett.* **92**, 097201 (2004).
- [25] A. Montanaro, Quantum speedup of Monte Carlo methods, *Proc. R. Soc. A* **471**, 20150301 (2015).
- [26] D. Poulin and P. Wocjan, Sampling from the thermal quantum Gibbs state and evaluating partition functions with a quantum computer, *Phys. Rev. Lett.* **103**, 220502 (2009).
- [27] A. N. Chowdhury, R. D. Somma, and Y. b. u. Subaşı, Computing partition functions in the one-clean-qubit model, *Phys. Rev. A* **103**, 032422 (2021).
- [28] A. Jackson, T. Kapourniotis, and A. Datta, Partition-function estimation: Quantum and quantum-inspired algorithms, *Phys. Rev. A* **107**, 012421 (2023).
- [29] K. Bharti, A. Cervera-Lierta, T. H. Kyaw, T. Haug, S. Alperin-Lea, A. Anand, M. Degroote, H. Heimonen, J. S. Kottmann, T. Menke, W.-K. Mok, S. Sim, L.-C. Kwek, and A. Aspuru-Guzik, Noisy intermediate-scale quantum algorithms, *Rev. Mod. Phys.* **94**, 015004 (2022).
- [30] A. Perdomo-Ortiz, S. E. Venegas-Andraca, and A. Aspuru-Guzik, A study of heuristic guesses for adiabatic quantum computation, *Quantum Inf. Process.* **10**, 33 (2011).
- [31] N. Chancellor, Modernizing quantum annealing using local searches, *New J. Phys.* **19**, 023024 (2017).
- [32] A. D. King *et al.*, Observation of topological phenomena in a programmable lattice of 1,800 qubits, *Nature (London)* **560**, 456 (2018).
- [33] M. Ohkuwa, H. Nishimori, and D. A. Lidar, Reverse annealing for the fully connected p -spin model, *Phys. Rev. A* **98**, 022314 (2018).
- [34] Y. Yamashiro, M. Ohkuwa, H. Nishimori, and D. A. Lidar, Dynamics of reverse annealing for the fully connected p -spin model, *Phys. Rev. A* **100**, 052321 (2019).
- [35] T. Albash and D. A. Lidar, Demonstration of a scaling advantage for a quantum annealer over simulated annealing, *Phys. Rev. X* **8**, 031016 (2018).
- [36] P. Hauke, H. G. Katzgraber, W. Lechner, H. Nishimori, and W. D. Oliver, Perspectives of quantum annealing:

- Methods and implementations, *Rep. Prog. Phys.* **83**, 054401 (2020).
- [37] A. Miessen, D. J. Egger, I. Tavernelli, and G. Mazzola, Benchmarking digital quantum simulations above hundreds of qubits using quantum critical dynamics, *PRX Quantum* **5**, 040320 (2024).
- [38] S. A. Guo, Y. K. Wu, J. Ye, L. Zhang, W. Q. Lian, R. Yao, Y. Wang, R. Y. Yan, Y. J. Yi, Y. L. Xu, B. W. Li, Y. H. Hou, Y. Z. Xu, W. X. Guo, C. Zhang, B. X. Qi, Z. C. Zhou, L. He, and L. M. Duan, A site-resolved two-dimensional quantum simulator with hundreds of trapped ions, *Nature (London)* **630**, 613 (2024).
- [39] Y. Lu, W. Chen, S. Zhang, K. Zhang, J. Zhang, J.-N. Zhang, and K. Kim, Implementing arbitrary Ising models with a trapped-ion quantum processor, *Phys. Rev. Lett.* **134**, 050602 (2025).
- [40] X. Qiu, P. Zoller, and X. Li, Programmable quantum annealing architectures with Ising quantum wires, *PRX Quantum* **1**, 020311 (2020).
- [41] S. Ebadi *et al.*, Quantum optimization of maximum independent set using Rydberg atom arrays, *Science* **376**, 1209 (2022).
- [42] A. Lucas, Ising formulations of many NP problems, *Front. Phys.* **2**, 5 (2014).
- [43] T. Albash and D. A. Lidar, Adiabatic quantum computation, *Rev. Mod. Phys.* **90**, 015002 (2018).
- [44] E. J. Crosson and D. A. Lidar, Prospects for quantum enhancement with diabatic quantum annealing, *Nat. Rev. Phys.* **3**, 466 (2021).
- [45] D. S. Wild, D. Sels, H. Pichler, C. Zanoci, and M. D. Lukin, Quantum sampling algorithms for near-term devices, *Phys. Rev. Lett.* **127**, 100504 (2021).
- [46] M. Vuffray, C. Coffrin, Y. A. Kharkov, and A. Y. Lokhov, Programmable quantum annealers as noisy Gibbs samplers, *PRX Quantum* **3**, 020317 (2022).
- [47] R. Shibukawa, R. Tamura, and K. Tsuda, Boltzmann sampling with quantum annealers via fast Stein correction, *Phys. Rev. Res.* **6**, 043050 (2024).
- [48] D. Hahn, M. Dupont, M. Schmitt, D. J. Luitz, and M. Bukov, Quantum many-body Jarzynski equality and dissipative noise on a digital quantum computer, *Phys. Rev. X* **13**, 041023 (2023).
- [49] L. Bassman Oftelie, K. Klymko, D. Liu, N. M. Tubman, and W. A. de Jong, Computing free energies with fluctuation relations on quantum computers, *Phys. Rev. Lett.* **129**, 130603 (2022).
- [50] D. Sherrington and S. Kirkpatrick, Solvable model of a spin-glass, *Phys. Rev. Lett.* **35**, 1792 (1975).
- [51] S. Kirkpatrick and D. Sherrington, Infinite-ranged models of spin-glasses, *Phys. Rev. B* **17**, 4384 (1978).
- [52] W. Barthel, A. K. Hartmann, M. Leone, F. Ricci-Tersenghi, M. Weigt, and R. Zecchina, Hiding solutions in random satisfiability problems: A statistical mechanics approach, *Phys. Rev. Lett.* **88**, 188701 (2002).
- [53] See Supplemental Material at <http://link.aps.org/supplemental/10.1103/8gmb-p619> for technical details and extended derivations, which also includes Refs. [54–64].
- [54] R. Monasson, R. Zecchina, S. Kirkpatrick, B. Selman, and L. Troyansky, Determining computational complexity from characteristic “phase transitions”, *Nature (London)* **400**, 133 (1999).
- [55] M. Mezard, G. Parisi, and R. Zecchina, Analytic and algorithmic solution of random satisfiability problems, *Science* **297**, 812 (2002).
- [56] W. G. Cochran, *Sampling Techniques* (Wiley, New York, 1977).
- [57] I. Bezáková, D. Štefankovič, V. V. Vazirani, and E. Vigoda, Accelerating simulated annealing for the permanent and combinatorial counting problems, *SIAM J. Comput.* **37**, 1429 (2008).
- [58] S. Trebst, D. A. Huse, and M. Troyer, Optimizing the ensemble for equilibration in broad-histogram Monte Carlo simulations, *Phys. Rev. E* **70**, 046701 (2004).
- [59] H. G. Katzgraber, S. Trebst, D. A. Huse, and M. Troyer, Feedback-optimized parallel tempering Monte Carlo, *J. Stat. Mech.* (2006) P03018.
- [60] P. Wocjan, C.-F. Chiang, D. Nagaj, and A. Abeyesinghe, Quantum algorithm for approximating partition functions, *Phys. Rev. A* **80**, 022340 (2009).
- [61] A. W. Harrow and A. Y. Wei, Adaptive quantum simulated annealing for Bayesian inference and estimating partition functions, in *Proceedings of the Fourteenth Annual ACM-SIAM Symposium on Discrete Algorithms* (Society for Industrial and Applied Mathematics, Philadelphia, USA, 2020), pp. 193–212.
- [62] S. Arunachalam, V. Havlicek, G. Nannicini, K. Temme, and P. Wocjan, Simpler (classical) and faster (quantum) algorithms for Gibbs partition functions, *Quantum* **6**, 789 (2022).
- [63] A. Cornelissen and Y. Hamoudi, A sublinear-time quantum algorithm for approximating partition functions, in *Proceedings of the 2023 Annual ACM-SIAM Symposium on Discrete Algorithms (SODA)* (Society for Industrial and Applied Mathematics, Philadelphia, USA, 2023), pp. 1245–1264.
- [64] T. de Lima Silva, L. Borges, and L. Aolita, Partition function estimation with a quantum coin toss, [arXiv:2411.17816](https://arxiv.org/abs/2411.17816).
- [65] H. Oberhofer, C. Dellago, and P. L. Geissler, Biased sampling of nonequilibrium trajectories: Can fast switching simulations outperform conventional free energy calculation methods?, *J. Phys. Chem. B* **109**, 6902 (2005).
- [66] D. Landau and K. Binder, *A Guide to Monte Carlo Simulations in Statistical Physics*, 5th ed. (Cambridge University Press, Cambridge, England, 2021).
- [67] S. Kullback and R. A. Leibler, On information and sufficiency, *Ann. Math. Stat.* **22**, 79 (1951).
- [68] G. B. Arous and A. Jagannath, Spectral gap estimates in mean field spin glasses, *Commun. Math. Phys.* **361**, 1 (2018).
- [69] R. Gheissari and A. Jagannath, On the spectral gap of spherical spin glass dynamics, *Ann. l’Inst. Henri Poincaré, Probab. Stat.* **55**, 756 (2019).
- [70] G. Passarelli, K.-W. Yip, D. A. Lidar, H. Nishimori, and P. Lucignano, Reverse quantum annealing of the p -spin model with relaxation, *Phys. Rev. A* **101**, 022331 (2020).
- [71] G. Passarelli, K.-W. Yip, D. A. Lidar, and P. Lucignano, Standard quantum annealing outperforms adiabatic reverse annealing with decoherence, *Phys. Rev. A* **105**, 032431 (2022).

- [72] A. Le and C.L. Baldwin, Adiabatic reverse annealing is robust to low-temperature decoherence, [arXiv:2511.16735](#).
- [73] H. Doga, B. Raubenolt, F. Cumbo, J. Joshi, F.P. DiFilippo, J. Qin, D. Blankenberg, and O. Shehab, A perspective on protein structure prediction using quantum computers, *J. Chem. Theory Comput.* **20**, 3359 (2024).
- [74] M. Cerezo, G. Verdon, H.-Y. Huang, L. Cincio, and P.J. Coles, Challenges and opportunities in quantum machine learning, *Nat. Comput. Sci.* **2**, 567 (2022).
- [75] H. Li, Z. Yao, and X. Qiu, Source code for “Quantum Annealing Algorithms for Estimating Ising Partition Functions”, Zenodo, [10.5281/zenodo.18753290](#) (2026).